

Contents

Multi-Perspective Advisory: Extensions, MCP, Skills, Memory & Scheduler (Phase 3 Focus)	1
Architecture Perspective	1
Security Perspective	1
Performance Perspective	1
Maintainability Perspective	1
Grok Build Integration Perspective	2
Overall: Proceed with confidence. These features make the panel a true “super-app” backend as users love in Codex/Claude desktops. Fix any validation gaps in revise.	2

Multi-Perspective Advisory: Extensions, MCP, Skills, Memory & Scheduler (Phase 3 Focus)

Perspectives: Architecture, Security, Performance, Maintainability, Grok Build Integration, User Experience (Backend)

Architecture Perspective

The extensions browser (CRUD for MCP/plugins/skills via CLI wrap + TOML) + MemoryService + Scheduler form a powerful “sovereign extensions layer”. Using event bus for notifications (e.g., MCP toggled) is elegant. Scheduler with tokio::time or simple cron fits the /loop / /goal from report perfectly. Memory injection (workspace MEMORY.md) aligns with cross-session persistence.

Recommendations: Make Config & Extensions Browser a trait-based service for future plugins. Scheduler should support persistent jobs (SQLite). MemoryService: Hybrid vector + BM25 if scaling (but keep simple first).

Score: 9/10

Security Perspective

MCP servers (stdio/HTTP) introduce external tool risks - validate commands/args strictly. Skills/hooks can run arbitrary scripts (PreToolUse deny powerful but audit carefully). Memory persistence: Sanitize any injected content. Scheduler: Prevent abuse (rate limits on recurring jobs).

Mitigations: Sandbox all MCP/spawned processes. User approval for new MCP installs. Audit hooks for dangerous patterns.

Score: 8/10 - Needs strict validation in impl.

Performance Perspective

CRUD on config (TOML rewrite + inspect) is lightweight. Scheduler: Use efficient tokio intervals; avoid busy loops. Memory: Efficient serialization. For many recurring jobs: Background task pool.

Recommendations: Cache parsed config. Batch MCP doctor checks.

Score: 9/10

Maintainability Perspective

Modular services good. Full CRUD + doctor mirroring CLI is maintainable. Docs for each extension type essential.

Recommendations: Versioned extension manifests. Automated tests for CRUD flows.

Score: 9.5/10

Grok Build Integration Perspective

Perfect: Wraps grok mcp/plugin/skills subcommands + inspect + /mcps toggle. Imagine bridge proxies /imagine. Import from Claude sessions. Matches ecosystem compatibility 100%.

Recommendations: Expose marketplace install via CLI. Live MCP toggle in event bus.

Score: 10/10

Overall: Proceed with confidence. These features make the panel a true “super-app” backend as users love in Codex/Claude desktops. Fix any validation gaps in revise.